

SOLIDWORKS API (.NET VBA C#) — Reference

(98/337): IEnvironment-IEquationMgr

SOLIDWORKS API Documentation (converted from CHM)

IEnvironment (Type)

IEnvironment Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEnvironment Interface

Allows you to analyze the text and geometry used to create a geometric tolerance symbol.
NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

Visual Basic Syntax

Visual Basic (Declaration)

```
Public Interface IEnvironment
```

Visual Basic (Usage)



```
Dim instance As IEnvironment
```

C#

```
public interface IEnvironment
```

C++/CLI

```
public interface class IEnvironment
```

Visual Basic for Applications (VBA) Syntax

See [Environment](#).

Example

[Analyze Text and Geometry in GTol Symbol \(C#\)](#)
[Analyze Text and Geometry in GTol Symbol \(VB.NET\)](#)
[Analyze Text and Geometry in GTol Symbol \(VBA\)](#)

Remarks

This interface can be useful if you have a note that contains a geometric tolerance symbol and you want to translate that note.

You can find the file containing the list of supported geometric tolerance symbols and their abbreviations in **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\gtol.sym**.

NOTE: All numeric values returned from IEnvironment are relative to a unit text height of 1.0; i.e., if a geometric tolerance symbol has a text height of 0.15, then multiply the numeric values returned by 0.15.

Accessors

[ISldWorks::GetEnvironment](#) and [ISldWorks::IGetEnvironment](#)

Access Diagram

[Environment](#)

See Also

[IEnvironment Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IEnvironment Interface Members

[See Also](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEnvironment Interface

The following tables list the members exposed by [IEnvironment](#).

Public Methods

Name	Description
 GetIsBound	Gets whether the specified geometric tolerance symbol is bound.
 GetSymArcs	Obsolete. Superseded by IEnvironment.GetSymArcs2 .
 GetSymArcs2	Gets an array of information about all arcs in the specified geometric tolerance symbol.
 GetSymCircles	Gets an array that defines all circles in the geometric tolerance symbol.
 GetSymEdgeCounts	Gets an array that contains the number of lines, arcs, circles, text strings, and triangles in the specified geometric tolerance symbol.
 GetSymLines	Gets an array that defines all lines in the specified geometric tolerance symbol.
 GetSymText	Gets an array containing the text associated with the specified geometric tolerance symbol.
 GetSymTextPoints	Gets an array that defines all text points in the specified geometric tolerance symbol.
 GetSymTriangles	Gets an array that defines all triangles in the specified geometric tolerance symbol.
 IGetSymArcs	Obsolete. Superseded by IEnvironment.GetSymArcs2 .
 IGetSymCircles	Gets an array that defines all circles in the specified geometric tolerance symbol.
 IGetSymEdgeCounts	Gets an array that contains the number of lines, arcs, circles, text strings, and triangles in the specified geometric tolerance symbol.
 IGetSymLines	Gets an array that defines all lines in the specified geometric tolerance symbol.
 IGetSymTextPoints	Gets an array that defines all text points in the specified geometric tolerance symbol.
 IGetSymTriangles	Gets an array that defines all triangles in the specified geometric tolerance symbol.

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See Also

[IEnvironment Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IEnvironment Interface Methods

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IEnvironment Interface

For a list of all members of this type, see [IEnvironment members](#).

Public Methods		
Name	Description	
 GetIsBound	Gets whether the specified geometric tolerance symbol is bound.	
 GetSymArcs	Obsolete. Superseded by IEnvironment.GetSymArcs2 .	
 GetSymArcs2	Gets an array of information about all arcs in the specified geometric tolerance symbol.	
 GetSymCircles	Gets an array that defines all circles in the geometric tolerance symbol.	
 GetSymEdgeCounts	Gets an array that contains the number of lines, arcs, circles, text strings, and triangles in the specified geometric tolerance symbol.	
 GetSymLines	Gets an array that defines all lines in the specified geometric tolerance symbol.	
 GetSymText	Gets an array containing the text associated with the specified geometric tolerance symbol.	
 GetSymTextPoints	Gets an array that defines all text points in the specified geometric tolerance symbol.	
 GetSymTriangles	Gets an array that defines all triangles in the specified geometric tolerance symbol.	
 IGetSymArcs	Obsolete. Superseded by IEnvironment.GetSymArcs2 .	
 IGetSymCircles	Gets an array that defines all circles in the specified geometric tolerance symbol.	
 IGetSymEdgeCounts	Gets an array that contains the number of lines, arcs, circles, text strings, and triangles in the specified geometric tolerance symbol.	
 IGetSymLines	Gets an array that defines all lines in the specified geometric tolerance symbol.	

 IGetSymTextPoints	Gets an array that defines all text points in the specified geometric tolerance symbol.
 IGetSymTriangles	Gets an array that defines all triangles in the specified geometric tolerance symbol.

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See Also

[IEnvironment Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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GetIsBound Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : GetIsBound Method (IEnvironment)

Gets whether the specified geometric tolerance symbol is bound.

.NET Syntax

Visual Basic (Declaration)

```
Sub GetIsBound( _
    ByVal SymId As System.String, _
    ByRef RetVal As System.Boolean _
)
```

Visual Basic (Usage)



```
Dim instance As IEnvironment
Dim SymId As System.String
Dim Retval As System.Boolean

instance.GetIsBound(SymId, Retval)
```

C#

```
void GetIsBound(
    System.string SymId,
    out System.bool RetVal
)
```

C++/CLI

```
void GetIsBound(
    & System.String^ SymId,
    [Out] System.bool RetVal
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

RetVal

True if the geometric tolerance symbol is bound, false if not

Visual Basic for Applications (VBA) Syntax

See [Environment::GetIsBound](#).

Example

See the [IEnvironment](#) examples.

Remarks

Bound controls the horizontal spacing of a geometric tolerance symbol within a row of text in notes and dimensions.

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSymArcs Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : GetSymArcs Method (IEnvironment)

Obsolete. Superseded by [IEnvironment::GetSymArcs2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymArcs( _  
    ByVal SymId As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEnvironment  
Dim SymId As System.String  
Dim value As System.Object  
  
value = instance.GetSymArcs(SymId)
```

C#

```
System.object GetSymArcs(  
    System.string SymId  
)
```

C++/CLI

```
System.Object^ GetSymArcs(  
& System.String^ SymId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where LibraryName and SymbolName are a [pre-defined SOLIDWORKS symbol](#) in the SOLIDWORKS text file **gtol.sym**, typically installed at **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english**.

NOTE: You must include the right- and left-angle brackets and separate LibraryName and SymbolName with a hyphen; for example, <MOD-DEG>.

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Environment::GetSymArcs](#).

Remarks

Each arc in the geometric tolerance symbol is defined by three points (center, arc start, and end). The size of the array returned is based on the number of arcs within this geometric tolerance symbol. You can determine this number using the return value arc count from [IEnvironment::GetSymEdgeCounts](#).

The format of return value is the following array of doubles (in this example, for the *i*th arc):

retval[9 * i + 0] = x coordinate of center point

retval[9 * i + 1] = y coordinate of center point

retval[9 * i + 2] = z coordinate of center point

retval[9 * i + 3] = x coordinate of arc start point

retval[9 * i + 4] = y coordinate of arc start point

retval[9 * i + 5] = z coordinate of arc start point

retval[9 * i + 6] = x coordinate of arc end point

retval[9 * i + 7] = y coordinate of arc end point

retval[9 * i + 8] = z coordinate of arc end point

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)
[IEnvironment::IGetSymArcs Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetSymArcs2 Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : GetSymArcs2 Method (IEnvironment)

Gets an array of information about all arcs in the specified geometric tolerance symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymArcs2( _  
    ByVal SymId As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEnvironment  
Dim SymId As System.String  
Dim value As System.Object  
  
value = instance.GetSymArcs2(SymId)
```

C#

```
System.object GetSymArcs2(  
    System.string SymId  
)
```

C++/CLI

```
System.Object^ GetSymArcs2(  
& System.String^ SymId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Environment::GetSymArcs2](#).

Example

See the [Environment](#) examples.

Remarks

Each arc in the geometric tolerance symbol is defined by three points (center, arc start, and arc end), whether the arc is filled, and whether the chord joining the start and end points is drawn. The size of the array returned is based on the number of arcs within this geometric tolerance symbol. You can determine this number by calling [Environment::GetSymEdgeCounts](#).

The return value contains the following array for each arc:

[x coordinate of center point,
y coordinate of center point,
z coordinate of center point,
x coordinate of start point,
y coordinate of start point,
z coordinate of start point,
x coordinate of end point,
y coordinate of end point,
z coordinate of end point,
whether arch is filled,
whether the chord joining the start and end points is drawn]

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)

Availability

SOLIDWORKS 2015 SP01, Revision Number 23.1

SOLIDWORKS API Help

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GetSymCircles Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : GetSymCircles Method (IEnvironment)

Gets an array that defines all circles in the geometric tolerance symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymCircles( _
    ByVal SymId As System.String _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEnvironment
Dim SymId As System.String
Dim value As System.Object

value = instance.GetSymCircles(SymId)
```

C#

```
System.object GetSymCircles(
    System.string SymId
)
```

C++/CLI

```
System.Object^ GetSymCircles(
    & System.String^ SymId
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Environment::GetSymCircles](#).

Remarks

Each circle in the geometric tolerance symbol is defined by its radius and center point. The size of the array returned is based on the number of circles within this symbol. You can determine this number using the return value circle count from [IEnvironment::GetSymEdgeCounts](#).

The format of return value is the following array of doubles (in this example, for the *i*th circle):

```
retval[4 * i] = radius
retval[4 * i + 1] = x coordinate of center point
retval[4 * i + 2] = y coordinate of center point
retval[4 * i + 3] = z coordinate of center point
```

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)
[IEnvironment::IGetSymCircles Method](#)

SOLIDWORKS API Help

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GetSymEdgeCounts Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : GetSymEdgeCounts Method (IEnvironment)

Gets an array that contains the number of lines, arcs, circles, text strings, and triangles in the specified geometric tolerance symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymEdgeCounts( _  
    ByVal SymId As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEnvironment  
Dim SymId As System.String  
Dim value As System.Object  
  
value = instance.GetSymEdgeCounts(SymId)
```

C#

```
System.object GetSymEdgeCounts(  
    System.string SymId  
)
```

C++/CLI

```
System.Object^ GetSymEdgeCounts(  
& System.String^ SymId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Environment::GetSymEdgeCounts](#).

Example

See the [IEnvironment](#) examples.

Remarks

Format of return information is the following array of short integers:

retval[0] = line count

retval[1] = arc count

retval[2] = circle count

retval[3] = text count

retval[4] = triangle count

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)
[IEnvironment::IGetSymEdgeCounts Method](#)
[IEnvironment::GetSymArcs Method](#)
[IEnvironment::GetSymCircles Method](#)
[IEnvironment::GetSymLines Method](#)
[IEnvironment::GetSymText Method](#)
[IEnvironment::GetSymTextPoints Method](#)
[IEnvironment::GetSymTriangles Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetSymLines Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : GetSymLines Method (IEnvironment)

Gets an array that defines all lines in the specified geometric tolerance symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymLines( _  
    ByVal SymId As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEnvironment  
Dim SymId As System.String  
Dim value As System.Object  
  
value = instance.GetSymLines(SymId)
```

C#

```
System.object GetSymLines(  
    System.string SymId  
)
```

C++/CLI

```
System.Object^ GetSymLines(  
& System.String^ SymId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Environment::GetSymLines](#).

Example

See the [IEnvironment](#) examples.

Remarks

Each line in the geometric tolerance symbol is defined by two points. The size of the array returned by this method is based on the number of lines within this geometric tolerance symbol. You can determine this number using the return value line count from [IEnvironment::GetSymEdgeCounts](#).

The format of return value is the following array of doubles (in this example, for the *i*th line):

retval[6 * i + 0] = x coordinate of first point

retval[6 * i + 1] = y coordinate of first point

retval[6 * i + 2] = z coordinate of first point

retval[6 * i + 3] = x coordinate of second point

retval[6 * i + 4] = y coordinate of second point

retval[6 * i + 5] = z coordinate of second point

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)
[IEnvironment::GetSymLines Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSymText Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : GetSymText Method (IEnvironment)

Gets an array containing the text associated with the specified geometric tolerance symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymText( _  
    ByVal SymId As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEnvironment  
Dim SymId As System.String  
Dim value As System.Object  
  
value = instance.GetSymText(SymId)
```

C#

```
System.object GetSymText(  
    System.string SymId  
)
```

C++/CLI

```
System.Object^ GetSymText(  
& System.String^ SymId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Environment::GetSymText](#).

Remarks

You can use [Environment::GetSymEdgeCounts](#) to determine how many text strings are in the specified geometric tolerance symbol. For example, passing <MOD-MMC> in SymID returns one text string ("M") and one circle.

The format of the return data is an array of strings. The size of the array returned is based on the number of text pieces in this symbol.

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSymTextPoints Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : GetSymTextPoints Method (IEnvironment)

Gets an array that defines all text points in the specified geometric tolerance symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymTextPoints( _  
    ByVal SymId As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEnvironment  
Dim SymId As System.String  
Dim value As System.Object  
  
value = instance.GetSymTextPoints(SymId)
```

C#

```
System.object GetSymTextPoints(  
    System.string SymId  
)
```

C++/CLI

```
System.Object^ GetSymTextPoints(  
& System.String^ SymId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Environment::GetSymTextPoints](#).

Remarks

The size of the array returned is based on the number of text pieces within this geometric tolerance symbol. You can determine this number using the return value text count from [IEnvironment::GetSymEdgeCounts](#).

The return value is an array of doubles as follows:

retval[0] x coordinate of text 1

retval[1] y coordinate of text 1

retval[2] z coordinate of text 1

retval[3] x coordinate of text 2

retval[4] y coordinate of text 2

retval[5] z coordinate of text 2

retval[(n*3-3)] X coordinate of text n

retval[(n*3-2)] Y coordinate of text n

retval[(n*3-1)] Z coordinate of text n

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)
[IEnvironment::IGetSymTextPoints Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSymTriangles Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : GetSymTriangles Method (IEnvironment)

Gets an array that defines all triangles in the specified geometric tolerance symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSymTriangles( _  
    ByVal SymId As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IEnvironment  
Dim SymId As System.String  
Dim value As System.Object  
  
value = instance.GetSymTriangles(SymId)
```

C#

```
System.object GetSymTriangles(  
    System.string SymId  
)
```

C++/CLI

```
System.Object^ GetSymTriangles(  
& System.String^ SymId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

Array (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Environment::GetSymTriangles](#).

Remarks

The size of the array returned by this method is based on the number of triangles in this geometric tolerance symbol. You can determine this number using the return value triangle count from [IEnvironment::GetSymEdgeCounts](#).

The format of the returned data is an array of doubles:

retval[0] x coordinate of triangle point 1

retval[1] y coordinate of triangle point 1

retval[2] z coordinate of triangle point 1

retval[3] x coordinate of triangle point 2

retval[4] y coordinate of triangle point 2

retval[5] z coordinate of triangle point 2

retval[6] x coordinate of triangle point 3

retval[7] y coordinate of triangle point 3

retval[8] z coordinate of triangle point 3

retval[9] Boolean returned as a double that describes if the triangle is filled

retval[10] describes the [line type](#) of the triangle

where this set of data repeats itself for each triangle in the specified geometric tolerance symbol.

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)
[IEnvironment::IGetSymTriangles Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSymArcs Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : IGetSymArcs Method (IEnvironment)

Obsolete. Superseded by [IEnvironment::GetSymArcs2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSymArcs( _  
    ByVal SymId As System.String _  
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IEnvironment
Dim SymId As System.String
Dim value As System.Double

value = instance.IGetSymArcs(SymId)
```

C#

```
System.double IGetSymArcs(
    System.string SymId
)
```

C++/CLI

```
System.double IGetSymArcs(
    & System.String^ SymId
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Each arc in the symbol is defined by three points (center, arc start, and end). The size of the array returned is based on the number of arcs within this geometric tolerance symbol. You can determine this number using the return value arc count from [IEnvironment::IGetSymEdgeCounts](#).

The format of return value is the following array of doubles (in this example, for the *i*th arc):

retval[9 * *i* + 0] = x coordinate of center point

retval[9 * *i* + 1] = y coordinate of center point

retval[9 * *i* + 2] = z coordinate of center point

retval[9 * *i* + 3] = x coordinate of arc start point

retval[9 * *i* + 4] = y coordinate of arc start point

retval[9 * *i* + 5] = z coordinate of arc start point

retval[9 * *i* + 6] = x coordinate of arc end point

retval[9 * *i* + 7] = y coordinate of arc end point

retval[9 * *i* + 8] = z coordinate of arc end point

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)
[IEnvironment::GetSymArcs Method](#)

Gets an array that defines all circles in the specified geometric tolerance symbol.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetSymCircles( _
    ByVal SymId As System.String _
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IEnvironment
Dim SymId As System.String
Dim value As System.Double

value = instance.IGetSymCircles(SymId)
```

C#

```
System.double IGetSymCircles(
    System.string SymId
)
```

C++/CLI

```
System.double IGetSymCircles(
    & System.String^ SymId
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Each circle in the symbol is defined by its radius and center point. The size of the array returned is based on the number of circles within this geometric tolerance symbol. You can determine this number using the return value circle count from [IEnvironment::IGetSymEdgeCounts](#).

The format of return value is the following array of doubles (in this example, for the *i*th circle):

```
retval[4 * i] = radius
retval[4 * i + 1] = x coordinate of center point
retval[4 * i + 2] = y coordinate of center point
retval[4 * i + 3] = z coordinate of center point
```

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)
[IEnvironment::IGetSymCircles Method](#)

IGetSymEdgeCounts Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : IGetSymEdgeCounts Method (IEnvironment)

Gets an array that contains the number of lines, arcs, circles, text strings, and triangles in the specified geometric tolerance symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSymEdgeCounts( _  
    ByVal SymId As System.String _  
) As System.Short
```

Visual Basic (Usage)



```
Dim instance As IEnvironment  
Dim SymId As System.String  
Dim value As System.Short
```

```
value = instance.IGetSymEdgeCounts(SymId)
```

C#

```
System.short IGetSymEdgeCounts(  
    System.string SymId  
)
```

C++/CLI

```
System.short IGetSymEdgeCounts(  
& System.String^ SymId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

- in-process, unmanaged C++: Pointer to an array of shorts (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Format of return information is the following array of short integers:

retval[0] = line count

retval[1] = arc count

retval[2] = circle count

retval[3] = text count

retval[4] = triangle count

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)
[IEnvironment::GetSymEdgeCounts Method](#)
[IEnvironment::IGetSymArcs Method](#)

IGetSymLines Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : IGetSymLines Method (IEnvironment)

Gets an array that defines all lines in the specified geometric tolerance symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSymLines( _  
    ByVal SymId As System.String _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IEnvironment  
Dim SymId As System.String  
Dim value As System.Double  
  
value = instance.IGetSymLines(SymId)
```

C#

```
System.double IGetSymLines(  
    System.string SymId  
)
```

C++/CLI

```
System.double IGetSymLines(  
& System.String^ SymId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Each line in the geometric tolerance symbol is defined by two points. The size of the array returned by this method is based on the number of lines within this geometric tolerance symbol. You can determine this number using the return value line count from [IEnvironment::IGetSymEdgeCounts](#).

The format of return value is the following array of doubles (in this example, for the *i*th line):

retval[6 * i + 0] = x coordinate of first point

retval[6 * i + 1] = y coordinate of first point

retval[6 * i + 2] = z coordinate of first point

retval[6 * i + 3] = x coordinate of second point

retval[6 * i + 4] = y coordinate of second point

retval[6 * i + 5] = z coordinate of second point

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)
[IEnvironment::GetSymLines Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSymTextPoints Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : IGetSymTextPoints Method (IEnvironment)

Gets an array that defines all text points in the specified geometric tolerance symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSymTextPoints(  
    ByVal SymId As System.String _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IEnvironment  
Dim SymId As System.String  
Dim value As System.Double  
  
value = instance.IGetSymTextPoints(SymId)
```

C#

```
System.double IGetSymTextPoints(  
    System.string SymId  
)
```

C++/CLI

```
System.double IGetSymTextPoints(  
& System.String^ SymId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks 20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

The size of the array returned is based on the number of text pieces within this geometric tolerance symbol. You can determine this number using the return value text count from [IEnvironment::IGetSymEdgeCounts](#).

The return value is an array of doubles as follows:

retval[0] x coordinate of text 1

retval[1] y coordinate of text 1

retval[2] z coordinate of text 1

retval[3] x coordinate of text 2

retval[4] y coordinate of text 2

retval[5] z coordinate of text 2

retval[(n*3-3)] X coordinate of text n

retval[(n*3-2)] Y coordinate of text n

retval[(n*3-1)] Z coordinate of text n

See Also

[IEnvironment Interface](#)
[IEnvironment Members](#)
[IEnvironment.GetSymTextPoints Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSymTriangles Method (IEnvironment)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEnvironment Interface](#) : IGetSymTriangles Method (IEnvironment)

Gets an array that defines all triangles in the specified geometric tolerance symbol.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSymTriangles( _  
    ByVal SymId As System.String _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IEnvironment  
Dim SymId As System.String  
Dim value As System.Double  
  
value = instance.IGetSymTriangles(SymId)
```

C#

```
System.double IGetSymTriangles(  
    System.string SymId  
)
```

C++/CLI

```
System.double IGetSymTriangles(  
& System.String^ SymId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymId

Name of the geometric tolerance symbol formatted as:

<LibraryName-SymbolName>

where *LibraryName* and *SymbolName* are in the SOLIDWORKS text file **C:\ProgramData\SolidWorks\SolidWorks20nn\lang\english\gtol.sym**.

NOTE: You must include the right- and left-angle brackets and separate *LibraryName* and *SymbolName* with a hyphen; for example, <MOD-DEG>.

Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The size of the array returned by this method is based on the number of triangles in this geometric tolerance symbol. You can determine this number using the return value triangle count from [IEnvironment::IGetSymEdgeCounts](#).

The format of the returned data is an array of doubles:

retval[0] x coordinate of triangle point 1

retval[1] y coordinate of triangle point 1

retval[2] z coordinate of triangle point 1

retval[3] x coordinate of triangle point 2

retval[4] y coordinate of triangle point 2

retval[5] z coordinate of triangle point 2

retval[6] x coordinate of triangle point 3

retval[7] y coordinate of triangle point 3

retval[8] z coordinate of triangle point 3

retval[9] Boolean returned as a double that describes if the triangle is filled

retval[10] describes the [line type](#) of the triangle

where this set of data repeats itself for each triangle in the specified geometric tolerance symbol.

See Also

[IEnvironment Interface](#)

[IEnvironment Members](#)

[IEnvironment::GetSymTriangles Method](#)

IEquationMgr (Type)

IEquationMgr Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEquationMgr Interface

Maintains a list of all of the existing equations in a model.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface IEquationMgr

Visual Basic (Usage)

Dim instance As [IEquationMgr](#)

C#

public interface IEquationMgr

C++/CLI

public interface class IEquationMgr

Visual Basic for Applications (VBA) Syntax

See [EquationMgr](#).

Example

[Automatically Solve Equations in Sequence \(C#\)](#)
[Automatically Solve Equations in Sequence \(VB.NET\)](#)
[Automatically Solve Equations in Sequence \(VBA\)](#)
[Get Equation Values \(C#\)](#)
[Get Equation Values \(VB.NET\)](#)
[Get Equation Values \(VBA\)](#)

Remarks

To add or edit an equation directly on an assembly component's model, you must edit the component's model in context.

Accessors

[IModelDoc2::GetEquationMgr](#)

Access Diagram

[EquationMgr](#)

See Also

[IEquationMgr Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IEquationMgr Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEquationMgr Interface

For a list of all members of this type, see [IEquationMgr members](#).

Public Methods

Name	Description
 Add	Obsolete. Superseded by IEquationMgr.Add2
 Add2	Adds an equation at the specified index.
 Add3	Adds an equation at the specified index for the specified configurations.
 ChangeSuppressionForAllConfigurations	Changes the suppression state of the specified equation in all configurations.
 ChangeSuppressionForConfiguration	Changes the suppression state of an equation in the specified configuration.
 Delete	Deletes the equation at the specified index.
 EvaluateAll	Evaluates all equations.
 GetConfigurationOption	Gets the configuration option for the equation at the specified index.
 GetCount	Gets the number of equations in the model.
 GetDisabledEquationCount	Gets the number of disabled equations in the model.
 IAdd3	Adds an equation at the specified index for the specified configurations.
 ISetEquationAndConfigurationOption	Modifies the equation at the specified index for the specified configurations.
 SetEquationAndConfigurationOption	Modifies the equation at the specified index for the specified configurations.
 UpdateValuesFromExternalEquationFile	Updates equations dependent on a linked equation file and ensures that the linked equation file exists and updates its current path, if necessary.

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See Also

[IEquationMgr Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IEquationMgr Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEquationMgr Interface

The following tables list the members exposed by [IEquationMgr](#).

Public Properties

Name	Description
 AngularEquationUnits	Gets or sets the angular units used in equations.
 AutomaticRebuild	Gets or sets whether to automatically rebuild after modifications.
 AutomaticSolveOrder	Gets or sets whether to automatically sequence equations in an order determined by SOLIDWORKS to produce accurate results.
 Disabled	Gets or sets whether to disable the specified equation in the model.
 Equation	Gets or sets the equation at the specified index.
 FilePath	Gets or sets the path for an exported equation text (.txt) file.
 GlobalVariable	Gets whether the equation at the specified index is a global variable.
 LinkToFile	Gets or sets whether the equation is linked to an exported equation text (.txt) file.
 Status	Gets the status of the last equation that was executed.
 Suppression	Obsolete as of SOLIDWORKS 2014 and later.
 Value	Gets the value of the equation at the specified index.

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Public Methods

Name	Description
------	-------------

	Add	Obsolete. Superseded by IEquationMgr.Add2
	Add2	Adds an equation at the specified index.
	Add3	Adds an equation at the specified index for the specified configurations.
	ChangeSuppressionForAllConfigurations	Changes the suppression state of the specified equation in all configurations.
	ChangeSuppressionForConfiguration	Changes the suppression state of an equation in the specified configuration.
	Delete	Deletes the equation at the specified index.
	EvaluateAll	Evaluates all equations.
	GetConfigurationOption	Gets the configuration option for the equation at the specified index.
	GetCount	Gets the number of equations in the model.
	GetDisabledEquationCount	Gets the number of disabled equations in the model.
	Add3	Adds an equation at the specified index for the specified configurations.
	ISetEquationAndConfigurationOption	Modifies the equation at the specified index for the specified configurations.
	SetEquationAndConfigurationOption	Modifies the equation at the specified index for the specified configurations.
	UpdateValuesFromExternalEquationFile	Updates equations dependent on a linked equation file and ensures that the linked equation file exists and updates its current path, if necessary.

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 See Also

[IEquationMgr Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IEquationMgr Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IEquationMgr Interface

For a list of all members of this type, see [IEquationMgr members](#).

 Public Properties

Name	Description
	AngularEquationUnits Gets or sets the angular units used in equations.
	AutomaticRebuild Gets or sets whether to automatically rebuild after modifications.
	AutomaticSolveOrder Gets or sets whether to automatically sequence equations in an order determined by SOLIDWORKS to produce accurate results.
	Disabled Gets or sets whether to disable the specified equation in the model.
	Equation Gets or sets the equation at the specified index.
	FilePath Gets or sets the path for an exported equation text (.txt) file.
	GlobalVariable Gets whether the equation at the specified index is a global variable.
	LinkToFile Gets or sets whether the equation is linked to an exported equation text (.txt) file.
	Status Gets the status of the last equation that was executed.
	Suppression Obsolete as of SOLIDWORKS 2014 and later.
	Value Gets the value of the equation at the specified index.

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 See Also

[IEquationMgr Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

Add Method (IEquationMgr)

[See Also](#)



.NET Syntax

Visual Basic (Declaration)

```
Function Add( _  
    ByVal Index As System.Integer, _  
    ByVal Equation As System.String _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim Index As System.Integer  
Dim Equation As System.String  
Dim value As System.Integer  
  
value = instance.Add(Index, Equation)
```

C#

```
System.int Add(  
    System.int Index,  
    System.string Equation  
)
```

C++/CLI

```
System.int Add(  
& System.int Index,  
& System.String^ Equation  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index of the new equation (-1 places it at the end of the list)
Equation
String containing the equation

Return Value

Index of the equation; -1 if there was an error

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::Add](#).

Remarks

The string specified for the equation argument must be formatted as if entered in the SOLIDWORKS Design user interface (that is, you must embed the names of the dimensions in double quotes). For example:

"N_SPOKES@CirPattern1" = "BARLENGTH@Sketch2" /10

You can also use the Visual Basic If function when specifying a model dimension in the equation argument. For example:

"D1@Extrude2" = (If("D1@Extrude3">20, 15, 6))+5

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::Delete Method](#)
[IEquationMgr::GetCount Method](#)
[IEquationMgr::Equation Property](#)
[IEquationMgr::Status Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Add2 Method (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : Add2 Method (IEquationMgr)

Adds an equation at the specified index.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function Add2( _  
    ByVal Index As System.Integer, _  
    ByVal Equation As System.String, _  
    ByVal Solve As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim Index As System.Integer  
Dim Equation As System.String  
Dim Solve As System.Boolean  
Dim value As System.Integer  
  
value = instance.Add2(Index, Equation, Solve)
```

C#

```
System.int Add2(  
    System.int Index,  
    System.string Equation,  
    System.bool Solve  
)
```

C++/CLI

```
System.int Add2(  
& System.int Index,  
& System.String^ Equation,  
& System.bool Solve  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the new equation (-1 places it at the end of the list)

Equation

String containing the equation (see **Remarks**)

Solve

True to solve the equation immediately; false otherwise (see **Remarks**)

Return Value

Index of the new equation if successfully added, -1 if an error

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::Add2](#).

Example

[Use If Function When Adding an Equation \(VBA\)](#)

[Add Equation and Evaluate All \(VB.NET\)](#)

[Add Equation and Evaluate All \(VBA\)](#)

[Add Equation and Evaluate All \(C#\)](#)

Remarks

To add an equation using the SOLIDWORKS Design user interface, you must embed the names of dimensions and global variables in double quotes:

- Global variable assignment:
"B" = 2
- Component equation:
"N_SPOKES@CirPattern" = "BARLENGTH@Sketch2" / 10
- Dimension equation that uses the Visual Basic If function:
"D1@Extrude2" = (If("D1@Extrude3">20, 15, 6))+5
- Dimension equation that sets a dimension to the current value:
"D1@Extrude2" =

NOTE: To add an equation using this method, you must specify Equation with the names of dimensions and global variables in double double quotes and the entire equation in double quotes. See the **Examples**.

If you call this method to add an equation whose left-hand side already exists in another equation, this method returns an error.

Setting the Solve argument to false delays evaluation of this equation, which enhances performance when there are many equations to solve. When Solve is set to false, the equation appears in the FeatureManager design tree only after calling [IEquationMgr::EvaluateAll](#) or [IModelDoc2::EditRebuild3](#).

To add equations to models having multiple configurations created in SOLIDWORKS 2014 or later, use [IEquationMgr::Add3](#) instead of this method.

When you add an equation, you are adding it to the list of relations. Thus, do not add an equation while traversing the equations in a model, because SOLIDWORKS could crash.

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::Delete Method](#)
[IEquationMgr::GetCount Method](#)
[IEquationMgr::Equation Property](#)
[IEquationMgr::Status Property](#)
[IEquationMgr::GetDisabledEquationCount Method](#)
[IEquationMgr::Disabled Property](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Add3 Method (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : Add3 Method (IEquationMgr)

Adds an equation at the specified index for the specified configurations.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function Add3( _  
    ByVal Index As System.Integer, _  
    ByVal Equation As System.String, _  
    ByVal Solve As System.Boolean, _  
    ByVal WhichConfigurations As System.Integer, _  
    ByVal ConfigNames As System.Object _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IEquationMgr  
Dim Index As System.Integer  
Dim Equation As System.String  
Dim Solve As System.Boolean  
Dim WhichConfigurations As System.Integer  
Dim ConfigNames As System.Object  
Dim value As System.Integer
```



```
value = instance.Add3(Index, Equation, Solve, WhichConfigurations, ConfigNames)
```

C#

```
System.int Add3(  
    System.int Index,  
    System.string Equation,  
    System.bool Solve,  
    System.int WhichConfigurations,  
    System.object ConfigNames  
)
```

C++/CLI

```
System.int Add3(  
& System.int Index,  
& System.String^ Equation,  
& System.bool Solve,  
& System.int WhichConfigurations,  
& System.Object^ ConfigNames  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the new equation (-1 places it at the end of the list)

Equation

String containing the equation (see **Remarks**)

Solve

True to solve the equation immediately; false to solve later (see **Remarks**)

WhichConfigurations

Configuration option as defined in [swInConfigurationOpts_e](#) (see **Remarks**)

ConfigNames

Array of the names of the configurations to which to add this equation; valid only if WhichConfigurations is set to swInConfigurationOpts_e.swSpecifyConfiguration, in which case, include the name of the current configuration in this array

Return Value

Index of the new equation if successfully added, -1 if error

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::Add3](#).

Example

[Add and Modify Equations \(VBA\)](#)
[Add and Modify Equations \(VB.NET\)](#)
[Add and Modify Equations \(C#\)](#)

Remarks

This method only works for parts having multiple configurations created in SOLIDWORKS 2014 or later. Use this method to add dimension equations, component equations, or global variable assignments to various configurations of models. If you don't have multiple configurations, then call [IEquationMgr::Add2](#), which does not support configurations and is slightly faster.

To add an equation using the SOLIDWORKS Design user interface, you must embed the names of dimensions and global variables in double quotes:

- Global variable assignment:
"B" = 2
- Component equation:
"N_SPOKES@CirPattern" = "BARLENGTH@Sketch2" /10
- Dimension equation that uses the Visual Basic If function:
"D1@Extrude2" = (If("D1@Extrude3">20, 15, 6))+5
- Dimension equation that sets a dimension to the current value:
"D1@Extrude2" =

NOTE: To add an equation:

- you must specify Equation with the names of dimensions and global variables in double double quotes and the entire equation in double quotes. See the Examples.
- directly to an assembly component's model, you must call [IAssemblyDoc::EditPart2](#) before calling this method.

You can add a dimension equation whose left-hand side already exists, but only if the existing equation is suppressed. Adding the equation again unsuppresses it. See [IEquationMgr::ChangeSuppressionForAllConfigurations](#) and [IEquationMgr::ChangeSuppressionForConfiguration](#).

Setting Solve to false delays evaluation of this equation, which enhances performance when there are many equations to solve. When Solve is set to false, the equation appears in the FeatureManager design tree only after calling [IEquationMgr::EvaluateAll](#) or [IModelDoc2::EditRebuild3](#).

When adding global variable assignments and component equations, WhichConfigurations must be set to [swInConfigurationOpts_e.swAllConfiguration](#).

If you add a global variable assignment that already exists, this method returns an error.

To modify an equation added by this method, call [IEquationMgr::SetEquationAndConfigurationOption](#).

When you add an equation, you are adding it to the list of relations. Thus, do not add an equation while traversing the equations in a model, because SOLIDWORKS could crash.

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::Delete Method \(\)](#)
[IEquationMgr::GetCount Method \(\)](#)
[IEquationMgr::Equation Property \(\)](#)
[IEquationMgr::Status Property \(\)](#)
[IEquationMgr::GetConfigurationOption Method \(\)](#)
[IEquationMgr::IAdd3 Method \(\)](#)
[IEquationMgr::ISetEquationAndConfigurationOption Method \(\)](#)
[IEquationMgr::Disabled Property \(\)](#)
[IEquationMgr::GetDisabledEquationCount Method \(\)](#)
[IEquationMgr::GlobalVariable Property \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AngularEquationUnits Property (IEquationMgr)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : AngularEquationUnits Property (IEquationMgr)

Gets or sets the angular units used in equations.

.NET Syntax

Visual Basic (Declaration)

Property AngularEquationUnits As System.Integer

Visual Basic (Usage)



```
Dim instance As IEquationMgr
Dim value As System.Integer

instance.AngularEquationUnits = value

value = instance.AngularEquationUnits
```

C#

```
System.int AngularEquationUnits {get; set;}
```

C++/CLI

```
property System.int AngularEquationUnits {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Angular units used in equations as defined in [swAngularEquationUnits_e](#)

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::AngularEquationUnits](#).

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr.Equation Property \(\)](#)

Availability

SOLIDWORKS 2006 SP5, Revision Number 14.5

SOLIDWORKS API Help

[Send comments on this topic.](#)

AutomaticRebuild Property (IEquationMgr)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : AutomaticRebuild Property (IEquationMgr)

Gets or sets whether to automatically rebuild after modifications.

.NET Syntax

Visual Basic (Declaration)

Property AutomaticRebuild As System.Boolean

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim value As System.Boolean  
  
instance.AutomaticRebuild = value  
value = instance.AutomaticRebuild
```

C#

```
System.bool AutomaticRebuild {get; set;}
```

C++/CLI

```
property System.bool AutomaticRebuild {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to automatically rebuild after modifications, false to not

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::AutomaticRebuild](#).

Remarks

This property corresponds to the **Automatic rebuild** checkbox in the Manage Equations dialog.

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr.Equation Property \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AutomaticSolveOrder Property (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : AutomaticSolveOrder Property (IEquationMgr)

Gets or sets whether to automatically sequence equations in an order determined by SOLIDWORKS to produce accurate results.

.NET Syntax

Visual Basic (Declaration)

Property AutomaticSolveOrder As System.Boolean

Visual Basic (Usage)



```
Dim instance As IEquationMgr
Dim value As System.Boolean

instance.AutomaticSolveOrder = value

value = instance.AutomaticSolveOrder
```

C#

```
System.bool AutomaticSolveOrder {get; set;}
```

C++/CLI

```
property System.bool AutomaticSolveOrder {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to automatically sequence equations in an order determined by SOLIDWORKS to produce accurate results and to prevent changing the order in which equations are solved, false to solve equations in the order in which they appear in the equation's Ordered View and to allow changing the order in which equations are solved

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::AutomaticSolveOrder](#).

Example

[Automatically Solve Equations in Sequence \(C#\)](#)
[Automatically Solve Equations in Sequence \(VB.NET\)](#)
[Automatically Solve Equations in Sequence \(VBA\)](#)

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::Equation Property \(\)](#)

Availability

SOLIDWORKS 2012 SP5, Revision 20.5

SOLIDWORKS API Help

[Send comments on this topic.](#)

ChangeSuppressionForAllConfigurations Method (IEquationMgr)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : ChangeSuppressionForAllConfigurations Method (IEquationMgr)

Changes the suppression state of the specified equation in all configurations.

.NET Syntax

Visual Basic (Declaration)

```
Function ChangeSuppressionForAllConfigurations( _
    ByVal Index As System.Integer, _
    ByVal State As System.Boolean _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr
Dim Index As System.Integer
Dim State As System.Boolean
Dim value As System.Integer

value = instance.ChangeSuppressionForAllConfigurations(Index, State)
```

C#

```
System.int ChangeSuppressionForAllConfigurations(
    System.int Index,
    System.bool State
)
```

C++/CLI

```
System.int ChangeSuppressionForAllConfigurations(
    & System.int Index,
    & System.bool State
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the target equation

State

True suppresses the equation, false unsuppresses it

Return Value

Index of the equation; -1 if error or if the equation was created in SOLIDWORKS 2014 or later

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::ChangeSuppressionForAllConfigurations](#).

See Also

[IEquationMgr Interface](#)

[IEquationMgr Members](#)

[IEquationMgr::ChangeSuppressionForConfiguration Method](#)

[IEquationMgr::GetCount Method](#)

[IEquationMgr::Status Property](#)

[IEquationMgr::Equation Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ChangeSuppressionForConfiguration Method (IEquationMgr)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : ChangeSuppressionForConfiguration Method (IEquationMgr)

Changes the suppression state of an equation in the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function ChangeSuppressionForConfiguration( _
    ByVal Index As System.Integer, _
```

```

        ByVal ConfigName As System.String, _
        ByVal State As System.Boolean _
    ) As System.Integer

```

Visual Basic (Usage)



```

Dim instance As IEquationMgr
Dim Index As System.Integer
Dim ConfigName As System.String
Dim State As System.Boolean
Dim value As System.Integer

value = instance.ChangeSuppressionForConfiguration(Index, ConfigName, State)

```

C#

```

System.int ChangeSuppressionForConfiguration(
    System.int Index,
    System.string ConfigName,
    System.bool State
)

```

C++/CLI

```

System.int ChangeSuppressionForConfiguration(
    & System.int Index,
    & System.String^ ConfigName,
    & System.bool State
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the target equation

ConfigName

Configuration in which to suppress the equation is (Nothing or null for the current configuration)

State

True suppresses the equation, false unsuppresses it

Return Value

Index of the equation; -1 if error or if the equation is created in SOLIDWORKS 2014 or later

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::ChangeSuppressionForConfiguration](#).

See Also

[IEquationMgr Interface](#)

[IEquationMgr Members](#)

[IEquationMgr::ChangeSuppressionForAllConfigurations Method](#)

[IEquationMgr::GetCount Method](#)

[IEquationMgr::Status Property](#)

[IEquationMgr::Equation Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Delete Method (IEquationMgr)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : Delete Method (IEquationMgr)

Deletes the equation at the specified index.

.NET Syntax

Visual Basic (Declaration)

```
Function Delete( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim Index As System.Integer  
Dim value As System.Integer  
  
value = instance.Delete(Index)
```

C#

```
System.int Delete(  
    System.int Index  
)
```

C++/CLI

```
System.int Delete(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the equation to delete

Return Value

0 if the equation is deleted; -1 if there was an error

Visual Basic for Applications (VBA) Syntax

See [EquationMgr.Delete](#).

Example

[Add Equation and Evaluate All \(C#\)](#)
[Add Equation and Evaluate All \(VB.NET\)](#)
[Add Equation and Evaluate All \(VBA\)](#)

Remarks

When you delete an equation, you are removing it from the list of relations. Thus, do not delete an equation while traversing the equations in a model, because SOLIDWORKS could crash.

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr.Add2 Method](#)
[IEquationMgr.Add3 Method](#)
[IEquationMgr.GetCount Method](#)
[IEquationMgr.Equation Property](#)
[IEquationMgr.Disabled Property](#)
[IEquationMgr.GetDisabledEquationCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Disabled Property (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : Disabled Property (IEquationMgr)

Gets or sets whether to disable the specified equation in the model.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property Disabled( _  
    ByVal Index As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim Index As System.Integer  
Dim value As System.Boolean  
  
instance.Disabled(Index) = value  
  
value = instance.Disabled(Index)
```

C#

```
System.bool Disabled(  
    System.int Index  
) {get; set;}
```

C++/CLI

```
property System.bool Disabled {  
    System.bool get(System.int Index);  
    void set (System.int Index, System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
0-based index of the equation

Property Value

True to disable the specified equation, false to not

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::Disabled](#).

Example

[Disable Equation \(C#\)](#)
[Disable Equation \(VB.NET\)](#)
[Disable Equation \(VBA\)](#)

Remarks

When you disable an equation, you are removing it from the list of relations. Thus, do not disable an equation while traversing the equations in a model, because SOLIDWORKS could crash.

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::Add2 Method \(\)](#)
[IEquationMgr::Add3 Method \(\)](#)
[IEquationMgr::Delete Method \(\)](#)
[IEquationMgr::GetCount Method \(\)](#)
[IEquationMgr::GetDisabledEquationCount Method \(\)](#)
[IEquationMgr::Equation Property \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

Equation Property (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : Equation Property (IEquationMgr)

Gets or sets the equation at the specified index.

.NET Syntax

Visual Basic (Declaration)

```
Property Equation( _  
    ByVal Index As System.Integer _  
) As System.String
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim Index As System.Integer  
Dim value As System.String  
  
instance.Equation(Index) = value  
  
value = instance.Equation(Index)
```

C#

```
System.string Equation(  
    System.int Index  
) {get; set;}
```

C++/CLI

```
property System.String^ Equation {  
    System.String^ get(System.int Index);  
    void set (System.int Index, System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
0-based index of the equation

Property Value

String containing the equation

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::Equation](#).

Example

[Use If Function When Adding an Equation \(VBA\)](#)

[Disable Equation \(C#\)](#)

[Disable Equation \(VB.NET\)](#)

[Disable Equation \(VBA\)](#)

[Get Equation Values \(C#\)](#)

[Get Equation Values \(VB.NET\)](#)

[Get Equation Values \(VBA\)](#)

Remarks

The string set by this property must be formatted as if entered in the SOLIDWORKS Design user interface; that is, you must embed the names of the dimensions and global variables in double double quotes. To use the following examples to specify Equation, you must replace every double quote with double double quotes, and enclose the entire equation with double quotes.

Global variable assignment:

"B" = 2

Component equation:

"N_SPOKES@CirPattern1" = "BARLENGTH@Sketch2" /10

Dimension equation that uses the Visual Basic If function:

"D1@Extrude2" = (("D1@Extrude3">20, 15, 6))+5

Dimension equation that sets a dimension to the current value:

"D1@Extrude2" =

Dimension equation that modifies the right-hand side of an already existing dimension equation:

"D1@Extrude2" = 0.05

NOTE: To set an equation:

- directly to an assembly component's model, you must call [IAssemblyDoc::EditPart2](#) before calling this method.
- for specific configurations in a model with multiple configurations, use [IEquationMgr::SetEquationAndConfigurationOption](#).

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::Add2 Method](#)
[IEquationMgr::Delete Method](#)
[IEquationMgr::GetCount Method](#)
[IEquationMgr::EvaluateAll Method](#)
[IEquationMgr::Add3 Method](#)
[IEquationMgr::Disabled Property](#)
[IEquationMgr::GetDisabledEquationCount Method](#)
[IEquationMgr::GlobalVariable Property](#)
[IEquationMgr::AutomaticSolveOrder Property](#)
[IEquationMgr::AutomaticRebuild Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

EvaluateAll Method (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : EvaluateAll Method (IEquationMgr)

Evaluates all equations.

.NET Syntax

Visual Basic (Declaration)

```
Function EvaluateAll() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim value As System.Integer  
  
value = instance.EvaluateAll()
```

C#

```
System.int EvaluateAll()
```

C++/CLI

```
System.int EvaluateAll();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

-1 for both success and failure

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::EvaluateAll](#).

Example

[Use If Function When Adding an Equation \(VBA\)](#)
[Add Equation and Evaluate All \(VBA\)](#)
[Add Equation and Evaluate All \(VB.NET\)](#)
[Add Equation and Evaluate All \(C#\)](#)

Remarks

Use this method to solve all equations that were added by [IEquationMgr::Add2](#) or [IEquationMgr::Add3](#) where Solve is set to false.

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::GetCount Method](#)
[IEquationMgr::Equation Property](#)
[IEquationMgr::Status Property](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FilePath Property (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : FilePath Property (IEquationMgr)

Gets or sets the path for an exported equation text (.txt) file.

.NET Syntax

Visual Basic (Declaration)

Property FilePath As System.String

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim value As System.String
```

```
instance.FilePath = value  
value = instance.FilePath
```

C#

```
System.string FilePath {get; set;}
```

C++/CLI

```
property System.String^ FilePath {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Path for the exported equation text (.txt) file

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::FilePath](#).

Example

[Pack and Go Part and Linked Equation \(C#\)](#)
[Pack and Go Part and Linked Equation \(VB.NET\)](#)
[Pack and Go Part and Linked Equation \(VBA\)](#)

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr.LinkToFile Property](#)
[IEquationMgr.UpdateValuesFromExternalEquationFile Method \(\)](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetConfigurationOption Method (IEquationMgr)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : GetConfigurationOption Method (IEquationMgr)

Gets the configuration option for the equation at the specified index.

.NET Syntax

Visual Basic (Declaration)

```
Function GetConfigurationOption( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim Index As System.Integer  
Dim value As System.Integer  
  
value = instance.GetConfigurationOption(Index)
```

C#

```
System.int GetConfigurationOption(  
    System.int Index  
)
```

C++/CLI

```
System.int GetConfigurationOption(  
& System.int Index  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
0-based index of equation for which to get the configuration option

Return Value

Configuration option as defined in [swlnConfigurationOpts_e](#)

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::GetConfigurationOption](#).

Remarks

This method only works for documents created in SOLIDWORKS 2014 or later.

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr.Add3 Method \(\)](#)
[IEquationMgr.IAdd3 Method \(\)](#)
[IEquationMgr.Equation Property \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetCount Method (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : GetCount Method (IEquationMgr)

Gets the number of equations in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function GetCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim value As System.Integer  
  
value = instance.GetCount()
```

C#

```
System.int GetCount()
```

C++/CLI

```
System.int GetCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of equations

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::GetCount](#).

Example

[Use If Function When Adding an Equation \(VBA\)](#)
[Automatically Solve Equations in Sequence \(C#\)](#)
[Automatically Solve Equations in Sequence \(VB.NET\)](#)
[Automatically Solve Equations in Sequence \(VBA\)](#)
[Get Equation Values \(C#\)](#)
[Get Equation Values \(VB.NET\)](#)
[Get Equation Values \(VBA\)](#)

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr.Add2 Method](#)
[IEquationMgr.ChangeSuppressionForAllConfigurations Method](#)
[IEquationMgr.ChangeSuppressionForConfiguration Method](#)
[IEquationMgr.Delete Method](#)
[IEquationMgr.Equation Property](#)
[IEquationMgr.EvaluateAll Method](#)
[IEquationMgr.Add3 Method](#)
[IEquationMgr.Disabled Property](#)
[IEquationMgr.GetDisabledEquationCount Method](#)
[IEquationMgr.GlobalVariable Property](#)

Availability

GetDisabledEquationCount Method (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : GetDisabledEquationCount Method (IEquationMgr)

Gets the number of disabled equations in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function GetDisabledEquationCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr
Dim value As System.Integer

value = instance.GetDisabledEquationCount()
```

C#

```
System.int GetDisabledEquationCount()
```

C++/CLI

```
System.int GetDisabledEquationCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of disabled equations in the model

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::GetDisabledEquationCount](#).

Example

[Disable Equation \(C#\)](#)
[Disable Equation \(VB.NET\)](#)
[Disable Equation \(VBA\)](#)

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::Disabled Property \(\)](#)
[IEquationMgr::GetCount Method \(\)](#)
[IEquationMgr::Equation Property \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

GlobalVariable Property (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : GlobalVariable Property (IEquationMgr)

Gets whether the equation at the specified index is a global variable.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property GlobalVariable( _  
    ByVal Index As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim Index As System.Integer  
Dim value As System.Boolean  
  
value = instance.GlobalVariable(Index)
```

C#

```
System.bool GlobalVariable(  
    System.int Index  
) {get;}
```

C++/CLI

```
property System.bool GlobalVariable {  
    System.bool get(System.int Index);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the equation

Property Value

True if the equation at the specified index is a global variable, false if not

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::GlobalVariable](#).

Example

[Get Equation Values \(C#\)](#)
[Get Equation Values \(VB.NET\)](#)
[Get Equation Values \(VBA\)](#)

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::GetCount Method \(\)](#)
[IEquationMgr::Equation Property \(\)](#)
[IEquationMgr::Value Property \(\)](#)

Availability

SOLIDWORKS 2016 SP4, Revision Number 24.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

IAdd3 Method (IEquationMgr)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : IAdd3 Method (IEquationMgr)

Adds an equation at the specified index for the specified configurations.

.NET Syntax

Visual Basic (Declaration)

```
Function IAdd3( _  
    ByVal Index As System.Integer, _
```

```

    ByVal Equation As System.String, _
    ByVal Solve As System.Boolean, _
    ByVal WhichConfigurations As System.Integer, _
    ByVal ConfigCount As System.Integer, _
    ByRef ConfigNames As System.String _
) As System.Integer

```

Visual Basic (Usage)



```

Dim instance As IEquationMgr
Dim Index As System.Integer
Dim Equation As System.String
Dim Solve As System.Boolean
Dim WhichConfigurations As System.Integer
Dim ConfigCount As System.Integer
Dim ConfigNames As System.String
Dim value As System.Integer

```

```
value = instance.IAdd3(Index, Equation, Solve, WhichConfigurations, ConfigCount, ConfigNames)
```

C#

```

System.int IAdd3(
    System.int Index,
    System.string Equation,
    System.bool Solve,
    System.int WhichConfigurations,
    System.int ConfigCount,
    ref System.string ConfigNames
)

```

C++/CLI

```

System.int IAdd3(
    & System.int Index,
    & System.String^ Equation,
    & System.bool Solve,
    & System.int WhichConfigurations,
    & System.int ConfigCount,
    & System.String^% ConfigNames
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the new equation (-1 places it at the end of the list)

Equation

String containing the equation (see **Remarks**)

Solve

True to solve the equation immediately; false to solve it later (see **Remarks**)

WhichConfigurations

Configuration option as defined in [swlnConfigurationOpts_e](#) (see **Remarks**)

ConfigCount

Number of names in the ConfigNames array

ConfigNames

- in-process, in-process, unmanaged C++: Pointer to an array of strings containing configuration names
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

Index of the new equation, -1 if error

Remarks

This method only works for parts having multiple configurations created in SOLIDWORKS 2014 or later. Use this method to add dimension equations, component equations, or global variable assignments to various configurations of models. If you don't have multiple configurations, then call [IEquationMgr.Add2](#), which does not support configurations and is slightly faster.

To add an equation using the SOLIDWORKS Design user interface, you must embed the names of dimensions and global variables in double quotes:

- Global variable assignment:
"B" = 2
- Component equation:
"N_SPOKES@CirPattern" = "BARLENGTH@Sketch2" /10

- Dimension equation that uses the Visual Basic If function:
"D1@Extrude2" = (If("D1@Extrude3">20, 15, 6))+5
- Dimension equation that sets a dimension to the current value:
"D1@Extrude2" =

NOTE: To add an equation:

- you must specify Equation with the names of dimensions and global variables in double double quotes and the entire equation in double quotes. See the Examples.
- directly to an assembly component's model, you must call [IAssemblyDoc::EditPart2](#) before calling this method.

You can add a dimension equation whose left-hand side already exists, but only if the existing equation is suppressed. Adding the equation again unsuppresses it. See [IEquationMgr::ChangeSuppressionForAllConfigurations](#) and [IEquationMgr::ChangeSuppressionForConfiguration](#).

Setting Solve to false delays evaluation of this equation, which enhances performance when there are many equations to solve. When Solve is set to false, the equation appears in the FeatureManager design tree only after calling [IEquationMgr::EvaluateAll](#) or [IModelDoc2::EditRebuild3](#).

When adding global variable assignments and component equations, WhichConfigurations must be set to [swInConfigurationOptions.e.swAllConfiguration](#).

If you add a global variable assignment that already exists, this method returns an error.

To modify an equation added by this method, call [IEquationMgr::ISetEquationAndConfigurationOption](#).

When you add an equation, you are adding it to the list of relations. Thus, do not add an equation while traversing the equations in a model, because SOLIDWORKS could crash.

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::GetConfigurationOption Method \(\)](#)
[IEquationMgr::Delete Method \(\)](#)
[IEquationMgr::GetCount Method \(\)](#)
[IEquationMgr::Equation Property \(\)](#)
[IEquationMgr::Status Property \(\)](#)
[IEquationMgr::Disabled Property \(\)](#)
[IEquationMgr::GetDisabledEquationCount Method \(\)](#)
[IEquationMgr::GlobalVariable Property \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetEquationAndConfigurationOption Method (IEquationMgr)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : ISetEquationAndConfigurationOption Method (IEquationMgr)

Modifies the equation at the specified index for the specified configurations.

.NET Syntax

Visual Basic (Declaration)

```
Function ISetEquationAndConfigurationOption( _
    ByVal Index As System.Integer, _
    ByVal Equation As System.String, _
    ByVal WhichConfigurations As System.Integer, _
    ByVal ConfigCount As System.Integer, _
    ByRef ConfigNames As System.String _
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IEquationMgr
Dim Index As System.Integer
Dim Equation As System.String
Dim WhichConfigurations As System.Integer
Dim ConfigCount As System.Integer
Dim ConfigNames As System.String
Dim value As System.Integer
```

```
value = instance.ISetEquationAndConfigurationOption(Index, Equation, WhichConfigurations,
    ConfigCount, ConfigNames)
```



C#

```
System.int ISetEquationAndConfigurationOption(  
    System.int Index,  
    System.string Equation,  
    System.int WhichConfigurations,  
    System.int ConfigCount,  
    ref System.string ConfigNames  
)
```

C++/CLI

```
System.int ISetEquationAndConfigurationOption(  
& System.int Index,  
& System.String^ Equation,  
& System.int WhichConfigurations,  
& System.int ConfigCount,  
& System.String^% ConfigNames  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the equation to modify

Equation

String containing the modified equation (see **Remarks**)

WhichConfigurations

Configuration option as defined in [swInConfigurationOpts_e](#)

ConfigCount

Number of names in ConfigNames array

ConfigNames

- in-process, in-process, unmanaged C++: Pointer to an array of strings containing configuration names
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

0 if equation successfully added, -1 if error

Remarks

This method modifies only equations added using [IEquationMgr::IAdd3](#).

To add an equation using the SOLIDWORKS Design user interface, you must embed the names of dimensions and global variables in double quotes:

- Global variable assignment:
"B" = 2
- Component equation:
"N_SPOKES@CirPattern" = "BARLENGTH@Sketch2" /10
- Dimension equation that uses the Visual Basic If function:
"D1@Extrude2" = (If("D1@Extrude3">20, 15, 6))+5
- Dimension equation that sets a dimension to the current value:
"D1@Extrude2" =
- Dimension equation that modifies the right-hand side of an already existing dimension equation:
"D1@Extrude2" = 0.05

NOTES:

- To modify an equation:
 - you must specify Equation with the names of dimensions and global variables in double double quotes and the entire equation in double quotes. The examples for [IEquationMgr::Add3](#) show how to do this. [Global variables](#) cannot be set to current values using this method.
 - added directly to an assembly component's model, you must call [IAssemblyDoc::EditPart2](#) before calling this method.
- If you change the active configuration, then you must call [IModelDoc2::GetEquationMgr](#) again.
- If the model has just one configuration, then use [IEquationMgr::Equation](#).

See Also

[IEquationMgr Interface](#)

[IEquationMgr Members](#)

[IEquationMgr::SetEquationAndConfigurationOption Method \(\)](#)

[IEquationMgr::GetConfigurationOption Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : LinkToFile Property (IEquationMgr)

Gets or sets whether the equation is linked to an exported equation text (.txt) file.

.NET Syntax

Visual Basic (Declaration)

Property LinkToFile As System.Boolean

Visual Basic (Usage)



```
Dim instance As IEquationMgr
Dim value As System.Boolean

instance.LinkToFile = value

value = instance.LinkToFile
```

C#

```
System.bool LinkToFile {get; set;}
```

C++/CLI

```
property System.bool LinkToFile {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the equation is linked to an exported equation text (.txt) file, false if not

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::LinkToFile](#).

Example

```
Dim swApp As SldWorks.SldWorks
Dim swmodel As SldWorks.ModelDoc2

Option Explicit

Sub main()

    Set swApp = Application.SldWorks
    Dim equationMgr As SldWorks.EquationMgr
    Set swmodel = swApp.ActiveDoc
    Set equationMgr = swmodel.GetEquationMgr
    equationMgr.FilePath = 'E:\API_SR\equations2.txt'
    equationMgr.LinkToFile = True

End Sub
```

Example

[Pack and Go Part and Linked Equation \(C#\)](#)
[Pack and Go Part and Linked Equation \(VB.NET\)](#)
[Pack and Go Part and Linked Equation \(VBA\)](#)

Remarks

The setter of this property works only after you have set [IEquationMgr::FilePath](#).

See Also

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetEquationAndConfigurationOption Method (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : SetEquationAndConfigurationOption Method (IEquationMgr)

Modifies the equation at the specified index for the specified configurations.

.NET Syntax

Visual Basic (Declaration)

```
Function SetEquationAndConfigurationOption( _  
    ByVal Index As System.Integer, _  
    ByVal Equation As System.String, _  
    ByVal WhichConfigurations As System.Integer, _  
    ByVal ConfigNames As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim Index As System.Integer  
Dim Equation As System.String  
Dim WhichConfigurations As System.Integer  
Dim ConfigNames As System.Object  
Dim value As System.Integer  
  
value = instance.SetEquationAndConfigurationOption(Index, Equation, WhichConfigurations,  
ConfigNames)
```

C#

```
System.int SetEquationAndConfigurationOption(  
    System.int Index,  
    System.string Equation,  
    System.int WhichConfigurations,  
    System.object ConfigNames  
)
```

C++/CLI

```
System.int SetEquationAndConfigurationOption(  
& System.int Index,  
& System.String^ Equation,  
& System.int WhichConfigurations,  
& System.Object^ ConfigNames  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the equation to modify

Equation

String containing the modified equation (see **Remarks**)

WhichConfigurations

Configuration option as defined in [swInConfigurationOpts_e](#)

ConfigNames

Array of the names of the configurations to which to add this equation; valid only if WhichConfigurations is set to swInConfigurationOpts_e.swSpecifyConfiguration, in which case, include the name of the current configuration in this array

Return Value

Index of equation if successfully modified, -1 if error

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::SetEquationAndConfigurationOption](#).

Example

[Add and Modify Equations \(VBA\)](#)
[Add and Modify Equations \(VB.NET\)](#)
[Add and Modify Equations \(C#\)](#)

Remarks

This method modifies only equations added using [IEquationMgr::Add3](#).

To add an equation using the SOLIDWORKS Design user interface, you must embed the names of dimensions and global variables in double quotes:

- Global variable assignment:
"B" = 2
- Component equation:
"N_SPOKES@CirPattern" = "BARLENGTH@Sketch2" /10
- Dimension equation that uses the Visual Basic If function:
"D1@Extrude2" = (If("D1@Extrude3">20, 15, 6))+5
- Dimension equation that sets a dimension to the current value:
"D1@Extrude2" =
- Dimension equation that modifies the right-hand side of an already existing dimension equation:
"D1@Extrude2" = 0.05

NOTES:

- To modify an equation:
 - you must specify Equation with the names of dimensions and global variables in double double quotes and the entire equation in double quotes. The examples for [IEquationMgr::Add3](#) show how to do this. [Global variables](#) cannot be set to current values using this method.
 - added directly to an assembly component's model, you must call [IAssemblyDoc::EditPart2](#) before calling this method.
- If you change the active configuration, then you must call [IModelDoc2::GetEquationMgr](#) again.
- If the model has just one configuration, then use [IEquationMgr::Equation](#).

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::GetConfigurationOption Method \(\)](#)
[IEquationMgr::ISetEquationAndConfigurationOption Method \(\)](#)
[IEquationMgr::IAdd3 Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Status Property (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : Status Property (IEquationMgr)

Gets the status of the last equation that was executed.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Status As System.Integer

Visual Basic (Usage)



```
Dim instance As IEquationMgr
Dim value As System.Integer
```

```
value = instance.Status
```

C#

```
System.int Status {get;}
```

C++/CLI

```
property System.int Status {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

0-based index of the last equation to successfully execute; -1 if there was an error (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::Status](#).

Example

[Get Equation Values \(C#\)](#)
[Get Equation Values \(VB.NET\)](#)
[Get Equation Values \(VBA\)](#)

Remarks

If you call an IEquationMgr method or property that takes the index of an equation as an argument (e.g., [IEquationMgr::Value](#)) and call IEquationMgr::Status, then the return value is:

- index of the equation if the call is successful.
- -1 if there is an error.

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::Add2 Method](#)
[IEquationMgr::ChangeSuppressionForAllConfigurations Method](#)
[IEquationMgr::ChangeSuppressionForConfiguration Method](#)
[IEquationMgr::Delete Method](#)
[IEquationMgr::Equation Property](#)
[IEquationMgr::EvaluateAll Method](#)
[IEquationMgr::Add3 Method](#)
[IEquationMgr::GetCount Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

Suppression Property (IEquationMgr)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : Suppression Property (IEquationMgr)

Obsolete as of SOLIDWORKS 2014 and later.

.NET Syntax

Visual Basic (Declaration)

```
Property Suppression( _  
    ByVal Index As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim Index As System.Integer  
Dim value As System.Boolean
```

```
instance.Suppression(Index) = value
```

```
value = instance.Suppression(Index)
```

C#

```
System.bool Suppression(  
    System.int Index  
) {get; set;}
```

C++/CLI

```
property System.bool Suppression {  
    System.bool get(System.int Index);  
    void set (System.int Index, System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
0-based index of the equation

Property Value

True if the equation is suppressed, false if not

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::Suppression](#).

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)

Availability

SOLIDWORKS 2001Plus, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

UpdateValuesFromExternalEquationFile Method (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : UpdateValuesFromExternalEquationFile Method (IEquationMgr)

Updates equations dependent on a linked equation file and ensures that the linked equation file exists and updates its current path, if necessary.

.NET Syntax

Visual Basic (Declaration)

```
Function UpdateValuesFromExternalEquationFile() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim value As System.Boolean
```

```
value = instance.UpdateValuesFromExternalEquationFile()
```

C#

```
System.bool UpdateValuesFromExternalEquationFile()
```

C++/CLI

```
System.bool UpdateValuesFromExternalEquationFile();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the equations are updated, false if not

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::UpdateValuesFromExternalEquationFile](#).

Example

[Pack and Go Part and Linked Equation \(C#\)](#)
[Pack and Go Part and Linked Equation \(VB.NET\)](#)
[Pack and Go Part and Linked Equation \(VBA\)](#)

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::LinkToFile Property \(\)](#)
[IEquationMgr::FilePath Property \(\)](#)

Availability

SOLIDWORKS 2014 SP1, Revision Number 22.1

SOLIDWORKS API Help

[Send comments](#) on this topic.

Value Property (IEquationMgr)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IEquationMgr Interface](#) : Value Property (IEquationMgr)

Gets the value of the equation at the specified index.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Value( _  
    ByVal Index As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IEquationMgr  
Dim Index As System.Integer  
Dim value As System.Double
```

```
value = instance.Value(Index)
```

C#

```
System.double Value(  
    System.int Index  
) {get;}
```

C++/CLI

```
property System.double Value {  
    System.double get(System.int Index);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

0-based index of the equation

Property Value

Value or -1 if Index is out of range

Visual Basic for Applications (VBA) Syntax

See [EquationMgr::Value](#).

Example

[Use If Function When Adding an Equation \(VBA\)](#)
[Get Equation Values \(C#\)](#)
[Get Equation Values \(VB.NET\)](#)
[Get Equation Values \(VBA\)](#)

Remarks

See Also

[IEquationMgr Interface](#)
[IEquationMgr Members](#)
[IEquationMgr::Status Property](#)
[IEquationMgr::GetCount Method](#)
[IEquationMgr::Equation Property](#)
[IEquationMgr::GlobalVariable Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0